QE Boron Carbon Input And Output

Irma Syafitri
Personal Summary
Applied Physics, Osaka University

October 2026

1 Boron And Carbon Input File

Quantum ESPRESSO Input File for Boron Atom

Here I'm sharing the input and output files for the Quantum ESPRESSO calculations of Boron and Carbon atoms, feel free to try them out!

Listing 1: Input file For Boron (B)

```
&control
    calculation = 'scf'
    restart_mode = 'from_scratch'
    prefix       = 'boron_atom'
    pseudo_dir   = '/home/irmasya/software/qe-7.5/pseudo/'
    outdir       = './out/'
/
&system
    ibrav        = 1
    celldm(1)    = 10.0
    nat          = 1
    ntyp         = 1
    nspin        = 2
    tot_magnetization = 1.0
    ecutwfc      = 40.0
    ecutrho      = 160.0
/
&electrons
    conv_thr     = 1.0d-8
    mixing_beta  = 0.7
/
ATOMIC_SPECIES
 B  10.81  B-PBE.upf
ATOMIC_POSITIONS crystal
 B  0.00000000  0.00000000  0.00000000
K_POINTS gamma
```

Quantum ESPRESSO Input File for Carbon Atom

Listing 2: Input file For Carbon (C)

```
&control
    calculation = 'scf'
    restart_mode = 'from_scratch'
    prefix       = 'carbon_atom'
    pseudo_dir   = '/home/irmasya/software/qe-7.5/pseudo/'
    outdir       = './out/'
/
&system
    ibrav        = 1
    celldm(1)    = 10.0
    nat          = 1
    ntyp         = 1
    nspin        = 2
    tot_magnetization = 2.0
    ecutwfc      = 40.0
    ecutrho      = 160.0
/
&electrons
    conv_thr     = 1.0d-8
    mixing_beta  = 0.7
/
ATOMIC_SPECIES
 C  12.011  C.UPF
ATOMIC_POSITIONS crystal
 C  0.00000000  0.00000000  0.00000000
K_POINTS gamma
```

2. Boron Atom Output

Listing 3: Boron SCF Output

```
     iteration # 98     ecut=    40.00 Ry     beta= 0.70
     Davidson diagonalization with overlap
     ethr =  9.33E-09,  avg # of iterations =  4.0

     total cpu time spent up to now is        10.6 secs

     total energy                  =      -5.42420370 Ry
     estimated scf accuracy    <         0.00000649 Ry

     total magnetization         =       1.00 Bohr mag/cell
     absolute magnetization      =       1.04 Bohr mag/cell

     iteration # 99     ecut=    40.00 Ry     beta= 0.70
```

```
     Davidson diagonalization with overlap
     ethr =  9.33E-09,  avg # of iterations =  4.0

     total cpu time spent up to now is        10.7 secs

     total energy                  =      -5.42416707 Ry
     estimated scf accuracy    <         0.00000391 Ry

     total magnetization         =       1.00 Bohr mag/cell
     absolute magnetization      =       1.04 Bohr mag/cell

     iteration #100     ecut=    40.00 Ry     beta= 0.70
     Davidson diagonalization with overlap
     c_bands:  1 eigenvalues not converged
     ethr =  9.33E-09,  avg # of iterations = 11.5

     total cpu time spent up to now is        10.8 secs

     total energy                  =      -5.42412204 Ry
     estimated scf accuracy    <         0.00000768 Ry
```

3. Carbon Atom Output

Listing 4: Carbon SCF Output

```
     total cpu time spent up to now is        10.6 secs

     total energy                  =     -10.77571923 Ry
     estimated scf accuracy    <         0.00001492 Ry

     total magnetization         =       2.00 Bohr mag/cell
     absolute magnetization      =       2.00 Bohr mag/cell

     iteration # 91     ecut=    40.00 Ry     beta= 0.70
     Davidson diagonalization with overlap
     c_bands:  2 eigenvalues not converged
     ethr =  4.21E-10,  avg # of iterations = 11.5

     total cpu time spent up to now is        10.7 secs

     total energy                  =     -10.77572341 Ry
     estimated scf accuracy    <         0.00006440 Ry

     total magnetization         =       2.00 Bohr mag/cell
     absolute magnetization      =       2.00 Bohr mag/cell

     iteration # 92     ecut=    40.00 Ry     beta= 0.70
     Davidson diagonalization with overlap
```

```
     c_bands:  2 eigenvalues not converged
     ethr =  4.21E-10,  avg # of iterations = 11.5

     total cpu time spent up to now is        10.9 secs

     total energy                  =     -10.77572550 Ry
     estimated scf accuracy    <         0.00000945 Ry

     total magnetization         =      2.00 Bohr mag/cell
     absolute magnetization      =      2.00 Bohr mag/cell

     iteration # 93     ecut=    40.00 Ry     beta= 0.70
     Davidson diagonalization with overlap
     c_bands:  2 eigenvalues not converged
     ethr =  4.21E-10,  avg # of iterations = 11.5

     total cpu time spent up to now is        11.1 secs

     total energy                  =     -10.77572620 Ry
     estimated scf accuracy    <         0.00000178 Ry

     total magnetization         =      2.00 Bohr mag/cell
     absolute magnetization      =      2.00 Bohr mag/cell

     iteration # 94     ecut=    40.00 Ry     beta= 0.70
     Davidson diagonalization with overlap
     c_bands:  2 eigenvalues not converged
     ethr =  4.21E-10,  avg # of iterations = 11.5

     Magnetic moment per site  (integrated on atomic sphere
         of radius R)
     atom   1 (R=0.413)  charge=    3.9714  magn=     1.9856

     total cpu time spent up to now is        11.1 secs

     End of self-consistent calculation
```